

g2(0) cases

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(Dated: September 19, 2026)

abstract

We consider a negatively charged quantum dot (QD) with a resident electron embedded in a symmetric optical microcavity. The optically active transition couples the ground state, in which the QD hosts a single resident electron, to the negatively charged trion X^- with energy E_t , built of an electron singlet and a heavy hole. The energy of the fundamental cavity mode $\hbar\omega_0$ coincides with that of the trion transition, $E_t = \hbar\omega_0$, and the interaction with all other modes is neglected. The system is driven by a laser pulse of carrier frequency ω_L , detuned from the transition by $\Delta = \omega_L - \omega_0$, and is subject to a magnetic field B_x , perpendicular to the growth axis. We are interested in the non-resonant regime, in which the detuning exceeds both the spectral width Γ_L of the pump pulse and the width Γ of the cavity mode, $\omega_L - \omega_0 \gtrsim \Gamma_L, \Gamma$, as sketched in Fig. 1. As the pump pulse carries a large number of photons, this separation of scales allows to describe it as a classical field with the Rabi frequency $\Omega_R(t)$, whereas the cavity mode, which is not directly fed by the laser, contains only a few photons and therefore has to be retained at the operator level.

Within the rotating wave approximation the Hamiltonian of the system, which accounts for the coupling to the cavity mode, for the interaction with the driving laser field and for the interaction with the magnetic field, reads (hereafter $\hbar = 1$)

$$\begin{aligned} \hat{H} = & E_t \sum_{\sigma=\uparrow,\downarrow} \hat{d}_\sigma^\dagger \hat{d}_\sigma + \omega_0 \sum_{\lambda=\pm} \hat{b}_\lambda^\dagger \hat{b}_\lambda \\ & + g \left(\hat{d}_\uparrow^\dagger \hat{a}_\uparrow \hat{b}_+ + \hat{d}_\downarrow^\dagger \hat{a}_\downarrow \hat{b}_- \right) + g^* \left(\hat{d}_\uparrow \hat{a}_\uparrow^\dagger \hat{b}_+^\dagger + \hat{d}_\downarrow \hat{a}_\downarrow^\dagger \hat{b}_-^\dagger \right) \\ & + \Omega_R(t) \sum_{\sigma=\uparrow,\downarrow} \left(e^{-i\omega_L t} \hat{d}_\sigma^\dagger \hat{a}_\sigma + e^{i\omega_L t} \hat{d}_\sigma \hat{a}_\sigma^\dagger \right) \\ & + \mu_B g_e B_x \left(\hat{a}_\downarrow^\dagger \hat{a}_\uparrow + \hat{a}_\uparrow^\dagger \hat{a}_\downarrow \right) + \mu_B g_t B_x \left(\hat{d}_\downarrow^\dagger \hat{d}_\uparrow + \hat{d}_\uparrow^\dagger \hat{d}_\downarrow \right) \end{aligned} \quad (1)$$

where the last two terms describe the Zeeman interaction in the transverse field. Here $\hat{a}_\sigma^\dagger$ is the creation operator of the resident electron, where $\sigma = \uparrow, \downarrow$ denotes the spin state $|\pm 1/2\rangle$; $\hat{d}_\sigma^\dagger$ is the creation operator of the negatively charged trion, where σ denotes the heavy hole spin state $|\pm 3/2\rangle$; $\hat{b}_\pm^\dagger = (\hat{b}_H^\dagger \pm i\hat{b}_V^\dagger)/\sqrt{2}$ is the creation operator of a cavity photon with circular polarization $\sigma_\pm$, expressed through the linearly polarized modes H and V ; g is the effective QD–cavity coupling parameter; μ_B is the Bohr

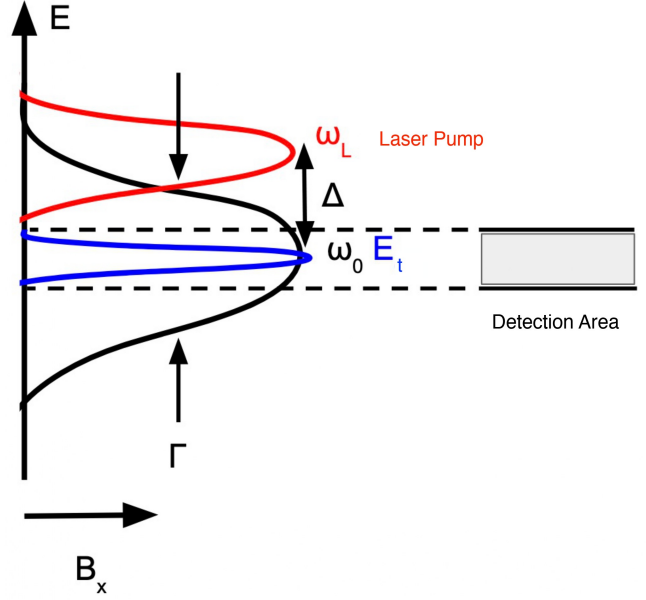


FIG. 1. Energy scheme of the system. The cavity mode of width Γ (black) is centered at ω_0 and is resonant with the trion transition E_t (blue). The pump pulse (red) has the carrier frequency ω_L and the spectral width Γ_L , and is detuned from the mode by Δ . The shaded box marks the spectral window of detection. The magnetic field B_x is applied perpendicular to the growth axis.

magneton, g_e and g_t are the in-plane g -factors of the resident electron and of the trion, respectively.

The optical selection rules fix the spin structure of the light–matter coupling: a σ_+ photon converts a resident electron $|+1/2\rangle$ into the trion with the heavy hole $|+3/2\rangle$, and a σ_- photon converts $|-1/2\rangle$ into $|-3/2\rangle$. Consequently, only the triads $(\hat{a}_\uparrow, \hat{d}_\uparrow, \hat{b}_+)$ and $(\hat{a}_\downarrow, \hat{d}_\downarrow, \hat{b}_-)$ are coupled, and the polarization of the emitted photon is rigidly tied to the spin state of the QD. The transverse field B_x , in turn, mixes the two spin branches, which is the mechanism that correlates the polarizations of the two emitted photons.

To describe the evolution of the system coupled to its environment, the Hamiltonian dynamics has to be supplemented by the dissipative terms of the Lindblad equation

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$$\partial_t \hat{\rho} = -i [\hat{H}, \hat{\rho}] + \sum_i \gamma_i \left(\hat{V}_i \hat{\rho} \hat{V}_i^\dagger - \frac{1}{2} \{ \hat{V}_i^\dagger \hat{V}_i, \hat{\rho} \} \right) \equiv \hat{\mathcal{L}} \hat{\rho} \quad (2)$$

where $\{\cdot, \cdot\}$ is the anticommutator, and γ_i and $\hat{V}_i$ are the rate and the jump operator of the i -th channel. Three dissipation channels are retained.

The first one is the escape of photons from the cavity mode into the environment, which is the process that delivers the photons to the detector. Its jump operator and rate are

$$\hat{V}_{\text{phot}} = \hat{b}_\pm, \quad \gamma_{\text{phot}} = \Gamma \quad (3)$$

where Γ is the width of the cavity mode.

The second channel is the pure dephasing of the optical transition induced by acoustic phonons, which is responsible for the damping of the Rabi oscillations. The corresponding jump operator is

$$\hat{V}_{\text{phon}} = \sum_{\sigma=\uparrow, \downarrow} \left(\hat{a}_\sigma^\dagger \hat{a}_\sigma - \hat{d}_\sigma^\dagger \hat{d}_\sigma \right), \quad \gamma_{\text{phon}} = \alpha \Omega_R^2 \quad (4)$$

The third channel is the dephasing of the resident electron spin. It is described by the operator

$$\hat{V}_{\text{spin } i} = \begin{pmatrix} \hat{a}_\uparrow^\dagger & \hat{a}_\downarrow^\dagger \end{pmatrix} \hat{\sigma}_i \begin{pmatrix} \hat{a}_\uparrow \\ \hat{a}_\downarrow \end{pmatrix} + \begin{pmatrix} \hat{d}_\uparrow^\dagger & \hat{d}_\downarrow^\dagger \end{pmatrix} \hat{\sigma}_i \begin{pmatrix} \hat{d}_\uparrow \\ \hat{d}_\downarrow \end{pmatrix} \quad (5)$$

where $\hat{\sigma}_i$ with $i = x, y, z$ are the Pauli matrices acting in the spin space of the resident electron and of the trion. Since the magnetic field is directed along x , the operators $\hat{V}_{\text{spin } y}$ and $\hat{V}_{\text{spin } z}$ relax the transverse spin component, whereas $\hat{V}_{\text{spin } x}$ adds a pure dephasing of the longitudinal ones. The three rates are therefore expressed through the longitudinal and transverse spin relaxation times T_1 and T_2 as

$$\gamma_{\text{spin } x} = \frac{1}{T_1}, \quad \gamma_{\text{spin } y} = \gamma_{\text{spin } z} = \frac{1}{T_2} \quad (6)$$

This channel destroys the coherence between the two spin branches and therefore directly degrades the polarization entanglement of the emitted photon pair.

Equation (2) is solved with the initial condition set by the state of the system before the arrival of the pump pulse, when the cavity mode is empty and the QD hosts only the resident electron. In the transverse magnetic field the eigenstates of the electron spin are the symmetric and antisymmetric superpositions $(|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$ split by the energy $2\mu_B g_e B_x$, and their populations follow the Boltzmann distribution at the lattice temperature T . The initial density matrix therefore reads

$$\begin{aligned} \hat{\rho}_0 = & \frac{f(B_x)}{2} \left(\hat{a}_\uparrow^\dagger + \hat{a}_\downarrow^\dagger \right) |0\rangle \langle 0| \left(\hat{a}_\uparrow + \hat{a}_\downarrow \right) \\ & + \frac{f(-B_x)}{2} \left(\hat{a}_\uparrow^\dagger - \hat{a}_\downarrow^\dagger \right) |0\rangle \langle 0| \left(\hat{a}_\uparrow - \hat{a}_\downarrow \right) \end{aligned} \quad (7)$$

where $f(B_x) = e^{\mu_B g_e B_x / T} / (e^{\mu_B g_e B_x / T} + e^{-\mu_B g_e B_x / T})$ and the temperature is measured in energy units.

The quantities measured in the experiment are the polarizations of the photons that reach the detector, so the observables are given by the projections of the full density matrix onto the sectors with a fixed number of emitted photons. Following Ref. [?], these sectors are conveniently expressed through the propagator $e^{\hat{\mathcal{L}}t}$ of the Lindbladian,

$$\hat{\rho}(t) = e^{\hat{\mathcal{L}}t} \hat{\rho}_0 \quad (8)$$

The one- and two-photon density matrices in the basis of the circular polarizations $\lambda = \pm$ then read

$$(\hat{\rho}_1)_{\lambda\lambda'} = \Gamma \int dt_1 \text{Tr} \left(\hat{b}_\lambda \hat{\rho}(t_1) \hat{b}_{\lambda'}^\dagger \right) \quad (9)$$

$$\begin{aligned} (\hat{\rho}_2)_{\lambda_1\lambda_2,\lambda'_1\lambda'_2} = & \Gamma^2 \int dt_1 dt_2 \text{Tr} \left(\hat{b}_{\lambda_2} e^{\hat{\mathcal{L}}(t_2-t_1)} \right. \\ & \left. \times \left[\hat{b}_{\lambda_1} \hat{\rho}(t_1) \hat{b}_{\lambda'_1}^\dagger \right] \hat{b}_{\lambda'_2}^\dagger \right) \end{aligned} \quad (10)$$

where Tr denotes the trace over all states except the photon states with $n = 0, 1, 2$. The probabilities of the one- and two-photon emission are the traces of these matrices,

$$P_1 = \text{Tr} \hat{\rho}_1, \quad P_2 = \text{Tr} \hat{\rho}_2 \quad (11)$$

As shown in Ref. [?], $\hat{\rho}_2$ contains two contributions of different nature: the resonant scattering of the laser photons, in which the whole dynamics stays coherent, and the re-excitation, in which the first photon actually leaves the cavity during the pulse and the remaining part of the pulse drives the QD again.

In our case the pulse is short, $\Gamma\tau \sim 1$, and in this regime the resonant scattering dominates [?]. Neglecting the escape of the photon from the mode during the pulse and the phonon-induced dephasing, the dynamics of this contribution is described by the following scheme

$$\begin{aligned} \hat{a}_\uparrow^\dagger |0\rangle + \hat{a}_\downarrow^\dagger |0\rangle & \xrightarrow{\Omega_R} \hat{d}_\uparrow^\dagger |0\rangle + \hat{d}_\downarrow^\dagger |0\rangle \\ & \xrightarrow{2\pi} \hat{b}_{+L}^\dagger \hat{d}_\uparrow^\dagger |0\rangle + \hat{b}_{-L}^\dagger \hat{d}_\downarrow^\dagger |0\rangle \\ & \longrightarrow \hat{b}_{+L}^\dagger \hat{b}_{+t}^\dagger \hat{a}_\uparrow^\dagger |0\rangle + \hat{b}_{-L}^\dagger \hat{b}_{-t}^\dagger \hat{a}_\downarrow^\dagger |0\rangle \\ & \xrightarrow{\text{spin relax}} \left(\hat{b}_{+L}^\dagger \hat{b}_{+t}^\dagger + \hat{b}_{-L}^\dagger \hat{b}_{-t}^\dagger \right) \left(\hat{a}_\uparrow^\dagger + \hat{a}_\downarrow^\dagger \right) |0\rangle \end{aligned} \quad (12)$$

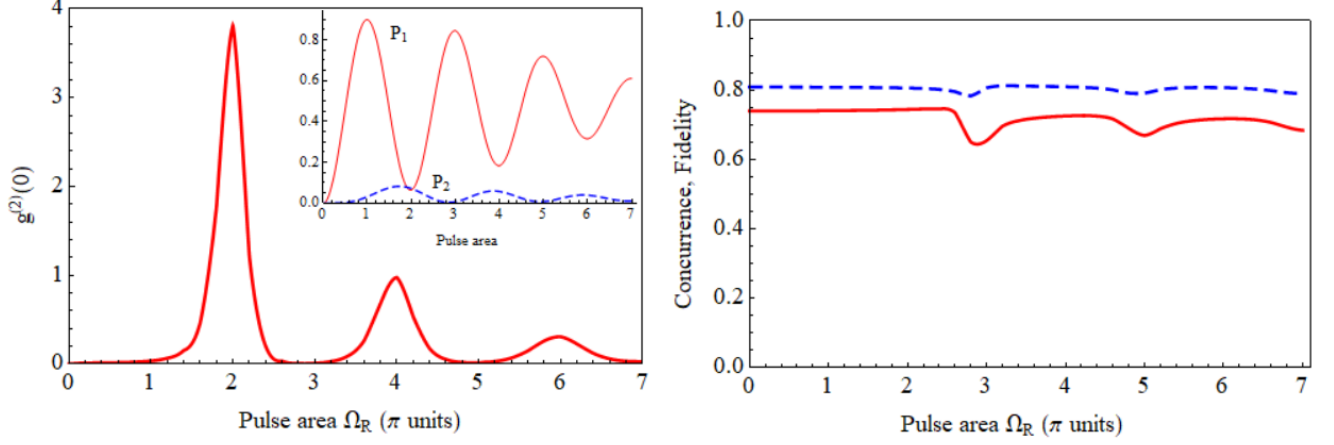


FIG. 2. (Left) Second-order correlation function at zero delay, $g^{(2)}(0)$, as a function of the pump pulse area. Inset: Probabilities of the one-photon, P_1 (solid), and two-photon, P_2 (dashed), emission as functions of the pump pulse area. (Right) Concurrence (solid) of the emitted two-photon polarization state and fidelity (dashed) to the Bell state $|\Phi^+\rangle$ as a function of the pump pulse area.

where $\hat{b}_{\lambda_L}^\dagger$ creates the resonantly scattered laser photon and $\hat{b}_{\lambda_t}^\dagger$ the photon emitted upon the trion recombination. The pulse first drives the resident electron into the trion, then a laser photon is resonantly scattered into the mode, and finally the trion recombines, returning the QD to its ground state.

As long as the trion has not recombined, the state of Eq. (12) carries a third index besides the two photon polarizations, namely the spin σ of the resident electron, and the selection rules lock all three of them to each other. The two-photon density matrix is accordingly a three-index object $(\hat{\rho}_2)_{\lambda_1\lambda_2\sigma,\lambda'_1\lambda'_2\sigma'}$, and the photons taken alone are entangled with the electron rather than with each other. The last step of Eq. (12) resolves this: the spin relaxes towards the in-plane thermal state $\hat{\rho}_{\text{spin}}$, the same one that the QD started from, irrespective of which pair has been emitted. Contracting the third index with it gives the density matrix of the photon pair alone,

$$\hat{\rho}_{\text{spin}} = \frac{1}{2} \begin{pmatrix} 1 & \eta \\ \eta & 1 \end{pmatrix}, \quad \eta = \tanh\left(\frac{\mu_B g_e B_x}{T}\right) \quad (13)$$

$$(\hat{\rho}_2)_{\lambda_1\lambda_2,\lambda'_1\lambda'_2} = \sum_{\sigma\sigma'} (\hat{\rho}_2)_{\lambda_1\lambda_2\sigma,\lambda'_1\lambda'_2\sigma'} (\hat{\rho}_{\text{spin}})_{\sigma'\sigma}$$

The contraction decouples the photonic degrees of freedom from the electronic ones. Since σ is locked to the photon polarizations, the populations, diagonal in σ , are untouched, whereas the coherences between the two branches are reduced by the factor η . The decoupling of the photonic and the electronic degrees of freedom is thus achieved at the expense of a partial loss of the polarization coherence, the residual spin polarization η setting an upper bound on the entanglement of the emitted pair.

Written as a density matrix with the thermal weights taken into account, the last line of Eq. (12) gives, to first order in $g\tau$ and in the basis $|++\rangle, |+-\rangle, |-+\rangle, |--\rangle$,

$$\hat{\rho}_2^{(2\pi)} \approx (g\tau)^2 \left(\frac{\Omega_R^2}{\Omega_R^2 + \Delta^2} \right)^2 \begin{pmatrix} 1 & 0 & 0 & \eta^2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \eta^2 & 0 & 0 & 1 \end{pmatrix} \quad (14)$$

Only the co-polarized states $|++\rangle$ and $|--\rangle$ are populated, and the entire dependence on the magnetic field and on the temperature enters through the coherence η^2 . The square appears because the spin coherence is picked up twice, once through the initial state (7) and once through the contraction (13). The factor $\Omega_R^2/(\Omega_R^2 + \Delta^2)$ shows how the detuning influences the two-photon yield.

Equation (14) is a mixture of two Bell states with thermal weights,

$$\hat{\rho}_2^{(2\pi)} \propto \frac{1+\eta^2}{2} |\Phi^+\rangle \langle \Phi^+| + \frac{1-\eta^2}{2} |\Phi^-\rangle \langle \Phi^-| \quad (15)$$

where $|\Phi^\pm\rangle = (|++\rangle \pm |--\rangle)/\sqrt{2}$. In the limit of a strong spin polarization, $\mu_B g_e B_x \gg T$, $\eta \rightarrow 1$ and only the maximally entangled $|\Phi^+\rangle$ survives, whereas for $\mu_B g_e B_x \ll T$ the two Bell states enter with equal weights and the pair is no longer entangled.

The quality of the generated state is characterized by two standard measures. The first one is the fidelity to the target Bell state,

$$F = \langle \Phi^+ | \hat{\rho}_2 | \Phi^+ \rangle \quad (16)$$

which exceeds 1/2 only for entangled states and therefore already serves as an entanglement witness. The second one is the concurrence, which quantifies the entanglement itself,

$$C = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4) \quad (17)$$

where $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$ are the eigenvalues of the matrix $\sqrt{\hat{\rho}_2} \hat{\rho}_2 \sqrt{\hat{\rho}_2}$, and $\hat{\rho}_2 = (\hat{\sigma}_y \otimes \hat{\sigma}_y) \hat{\rho}_2^* (\hat{\sigma}_y \otimes \hat{\sigma}_y)$ is the spin flipped density matrix, the complex conjugation being taken in the polarization basis. The concurrence vanishes for separable states and reaches unity for a maximally entangled one.

All the results presented below are obtained by a numerical solution of Eq. (2) with the following set of parameters: the cavity mode width $\Gamma = 50 \mu\text{eV}$, the QD-cavity coupling $g = 25 \mu\text{eV}$, the detuning $\Delta = 50 \mu\text{eV}$, the pulse duration $\tau = 13 \text{ ps}$, the phonon coupling constant $\alpha = 0.15 \text{ ps}$, the transverse magnetic field $B_x = 5 \text{ T}$ and the lattice temperature $T = 1 \text{ K}$. The pulse area $\theta = \int \Omega_R(t) dt$ is varied by changing the pump power at a fixed pulse duration, and is measured below in units of π .

The emission probabilities are shown in the inset of Fig. 2. The one-photon probability P_1 exhibits damped

Rabi oscillations with maxima at odd and minima at even multiples of π , the damping being caused by the phonon-induced dephasing. The two-photon probability P_2 oscillates in antiphase and remains an order of magnitude smaller, reaching its maxima close to even pulse areas.

This antiphase behaviour is directly reflected in the second-order correlation function, Fig. 2. At odd pulse areas $g^{(2)}(0)$ nearly vanishes, so that the source emits single photons, whereas at even ones it displays pronounced bunching peaks, the largest reaching $g^{(2)}(0) \approx 3.8$ at $\theta = 2\pi$.

The polarization state of the emitted pair is characterized in Fig. 2. Both the concurrence and the fidelity to the Bell state $|\Phi^+\rangle$ are only weakly sensitive to the pump power, staying close to 0.7 and 0.8, respectively, over the whole range of pulse areas, with shallow minima near the odd ones. The fidelity exceeding 1/2 confirms that the emitted pair is entangled for any excitation power.